

Accelerator Performance During the Beam Energy Scan II in 2019

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NAPAC'19, Lansing, MI, USA

09/02/2019

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Outline

Motivation for Beam Energy Scan

Challenges for Beam Energy Scan

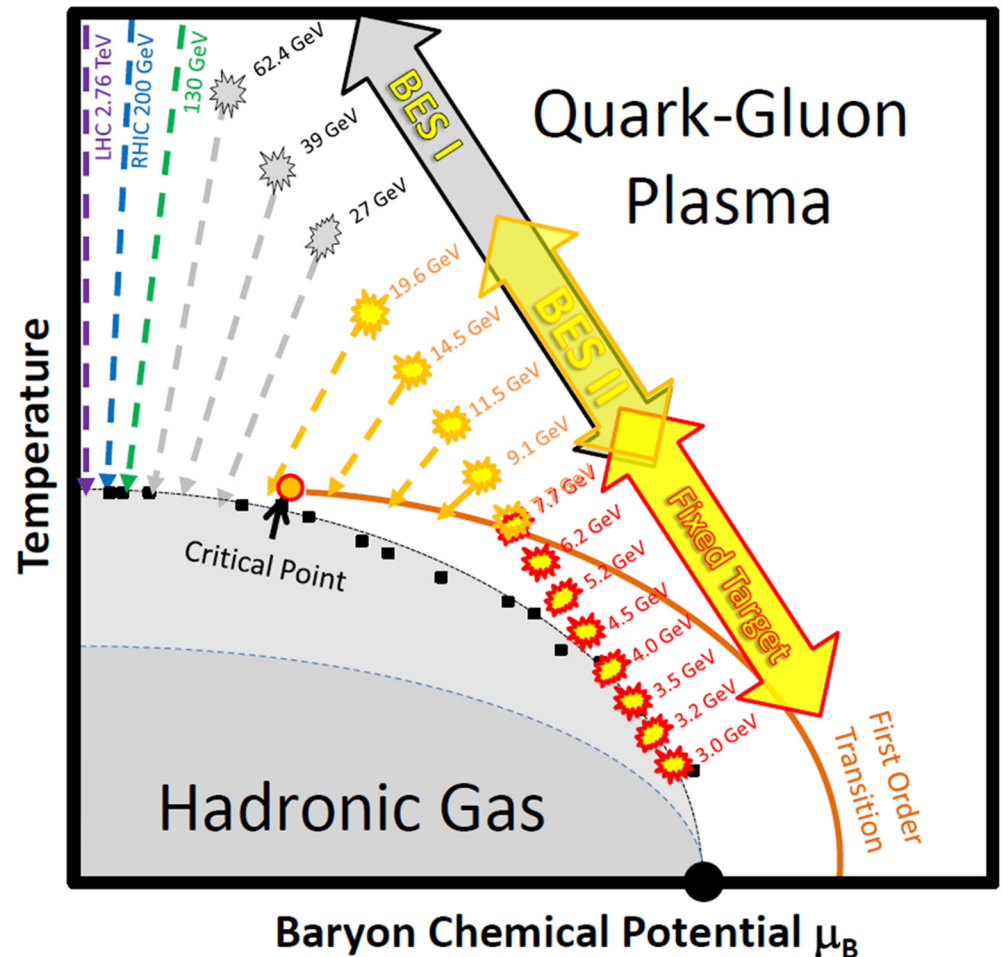
Accelerator performance in colliding mode

Accelerator performance in fixed target mode

Summary and outlook

Motivation for Beam Energy Scan

- To understand the transformation from hadronic gas to Quark-Gluon plasma (QGP).
- To locate a possible critical point of first-order phase transition.



Beam Energy Scan I & II

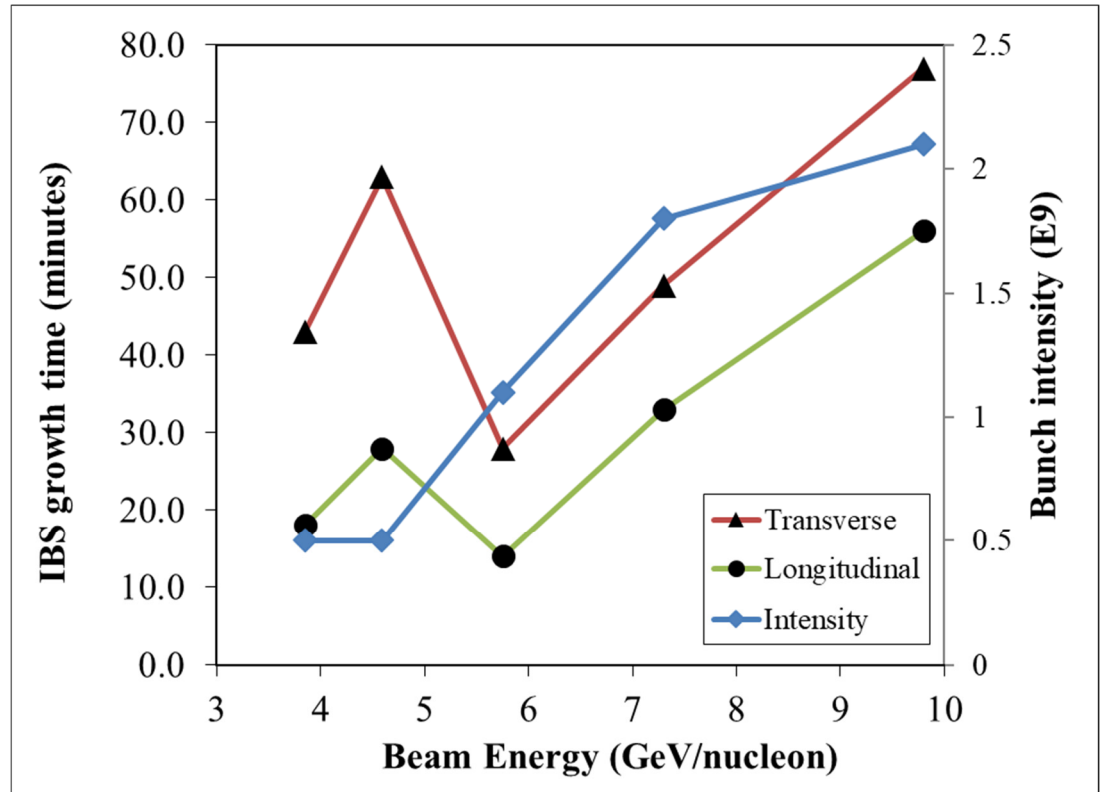
BES-I completed in 2014 improved understanding of many physics phenomena.

Transition between QGP and hadronic gas has not been understood due to limited statistics.

BES-II was planned (now ongoing) with luminosity improved by a factor of ~ 4 at the same beam energies as BES-I (3.85, 4.59, 5.75, 7.3 and 9.8 GeV/nucleon).

Challenge #1: Intra-beam scattering

- At BES-I/II beam energies, which are below the transition energy (~ 24 GeV/nucleon for Au beam), both longitudinal and transverse beam emittance grow rapidly due to intra-beam scattering (IBS).



LEReC cooling

- The Low Energy RHIC electron Cooling (LEReC), using a linear electron accelerator, is designed to combat the IBS effect by cooling RHIC ion beams, and therefore to improve luminosities at the two lowest beam energies.

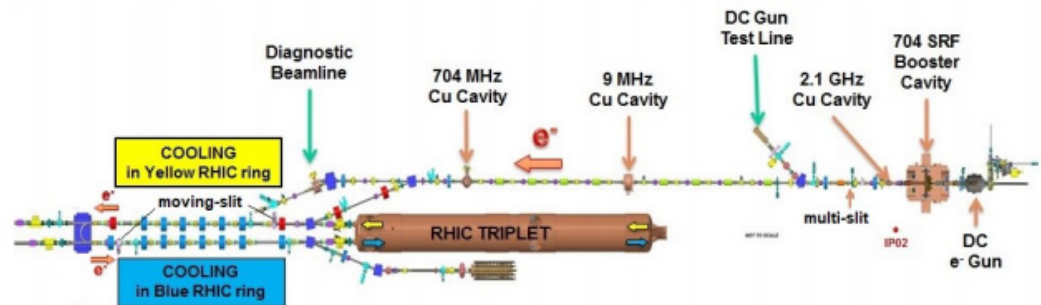
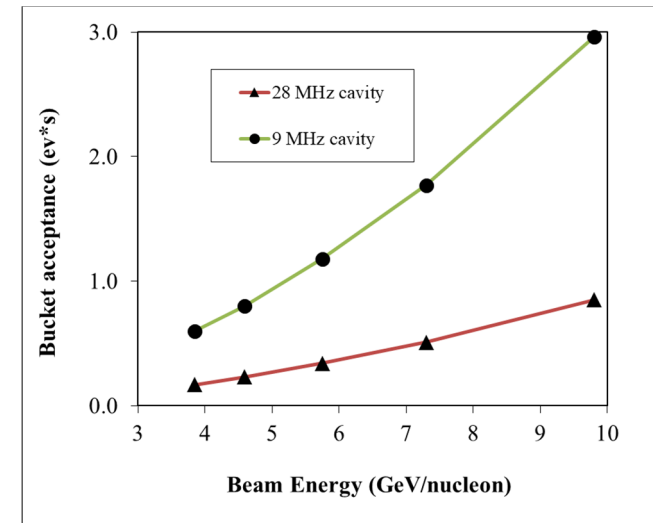
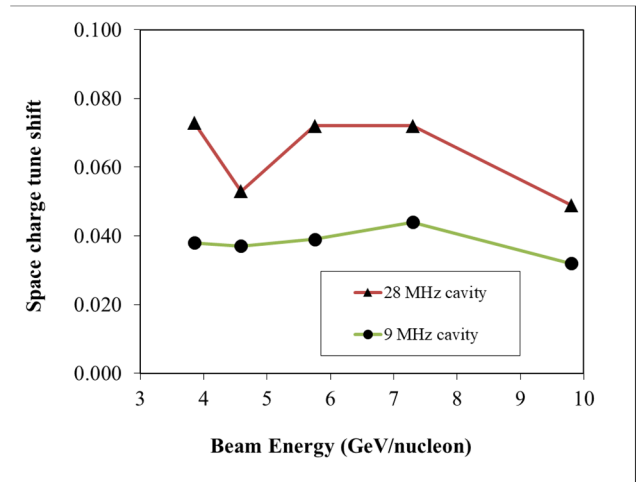


Figure 1: Schematics of Low Energy RHIC electron Cooling accelerator. Low energy electron beam is transported through a beamline with solenoids, merged into a cooling section in the Yellow RHIC ring first, turned around and merged into a cooling section in the Blue RHIC ring.

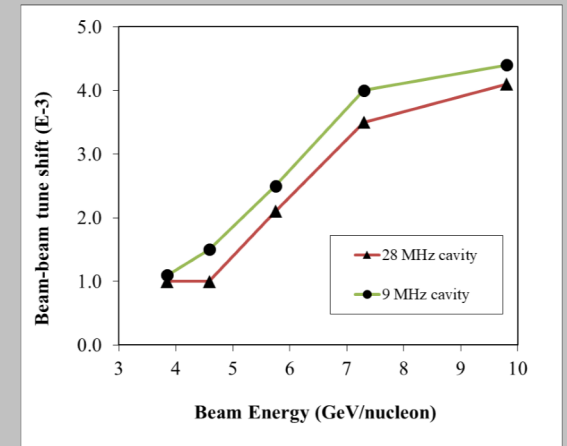
Challenges #2: Space charge

- At the BES-I/II beam energies, the ions in the beam experience a strong direct space charge force. The space charge force introduces incoherent and coherent tune shifts.
- The space charge effects are significantly reduced by using the new 9 MHz cavities due to reduced peak beam intensity.
- The 9 MHz cavities provide a larger bucket area than the 28 MHz cavities.

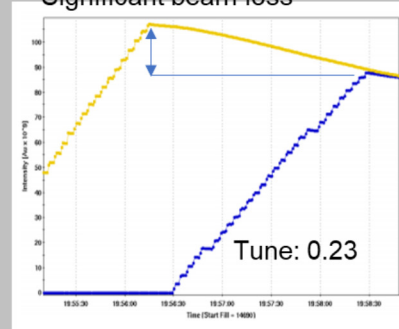


Challenges #3: beam-beam

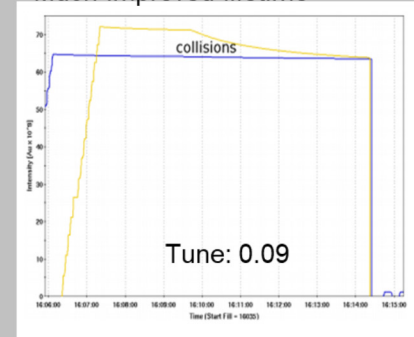
- Beam-beam is relative weak.
- Interplay of beam-beam and space charge hurts lifetime.
- Working point below 0.1 provides large tune space and alleviates beam loss.



Significant beam loss

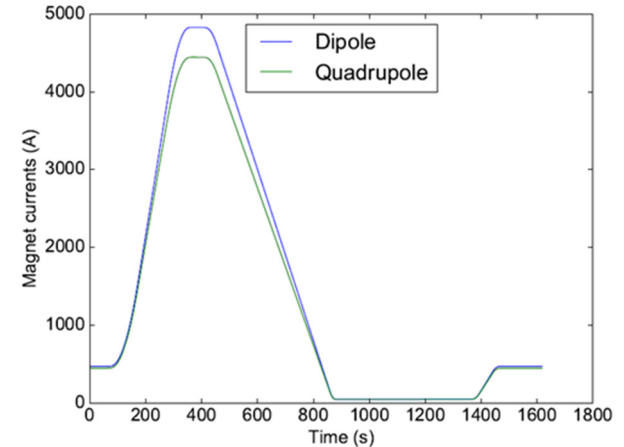


Much improved lifetime

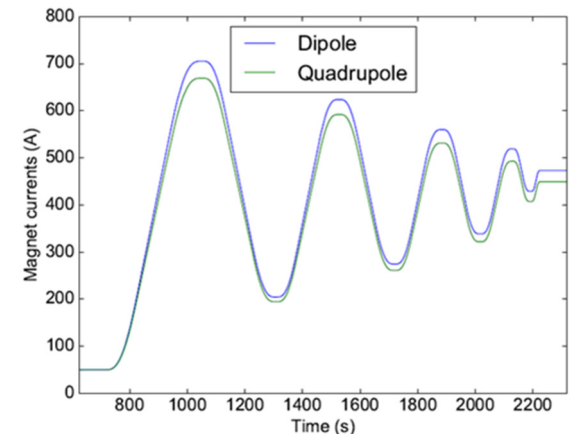


Challenge #4: persistent current in superconducting magnet

- Persistent currents in superconducting magnets introduce significant field errors.
- The decay of magnetic field errors causes the drifts of beam orbits, tunes and chromaticities.
- A demagnetization (wiggle) magnet cycle has been demonstrated and implemented to reduce field errors and its variation in time.



Regular cycle



Demagnetization cycle

Reduction of persistent current by demagnetization cycle

- Sextupole components in dipoles are reduced.
- Orbit, tune and chromaticity drifts are significantly reduced.

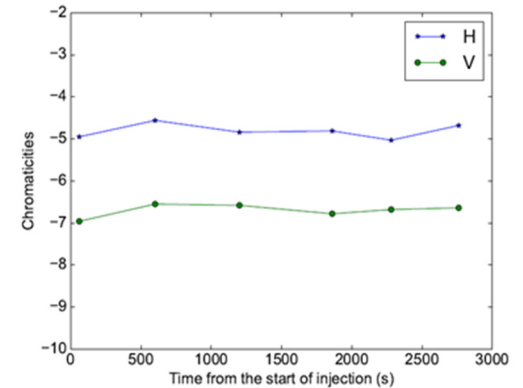
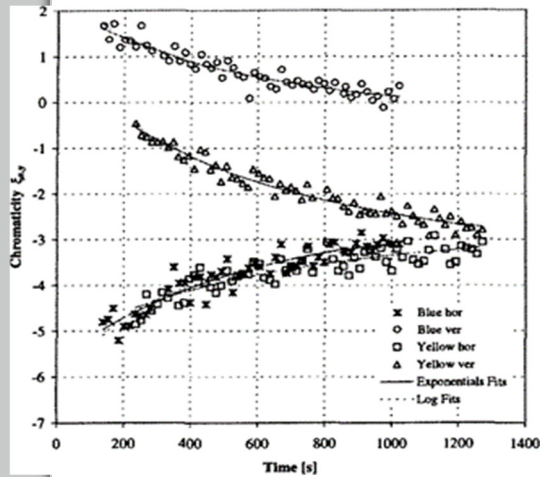
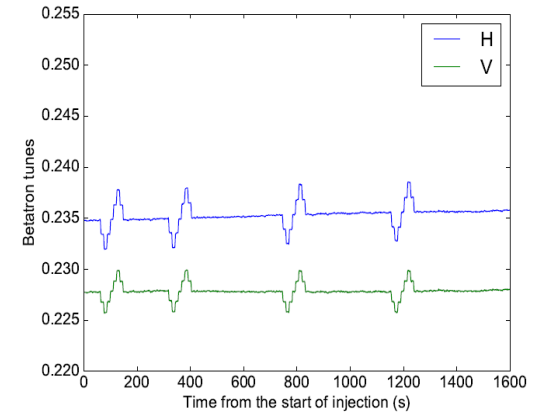
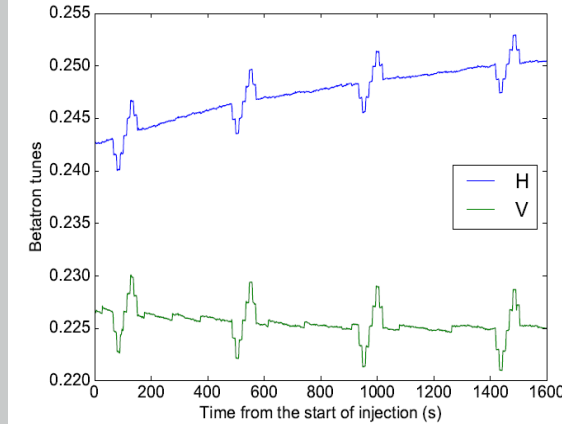
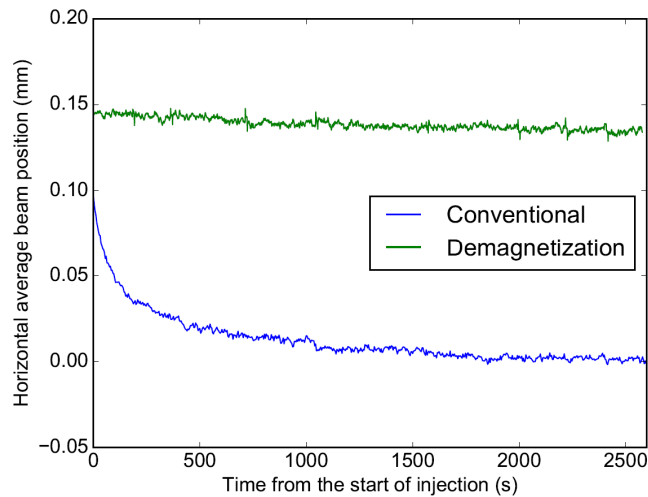


FIG. 8: The measured beam chromaticities after the demagnetization cycle.

Beam Energy Scan II (BES-II)

Beam Energy (GeV/nucleon)	$\sqrt{s_{NN}}$ (GeV)	μ_B (MeV)	Run Time	Number Events
9.8	19.6	205	4.5 weeks	400M
7.3	14.5	260	5.5 weeks	300M
5.75	11.5	315	5 weeks	230M
4.55 ★	9.1	370	9.5 weeks	160M
3.85 ★	7.7	420	12 weeks	100M
31.2	7.7 (FXT)	420	2 days	100M
19.5	6.2 (FXT)	487	2 days	100M
13.5	5.2 (FXT)	541	2 days	100M
9.8	4.5 (FXT)	589	2 days	100M
7.3	3.9 (FXT)	633	2 days	100M
5.75	3.5 (FXT)	666	2 days	100M
4.55	3.2 (FXT)	699	2 days	100M
3.85	3.0 (FXT)	721	2 days	100M

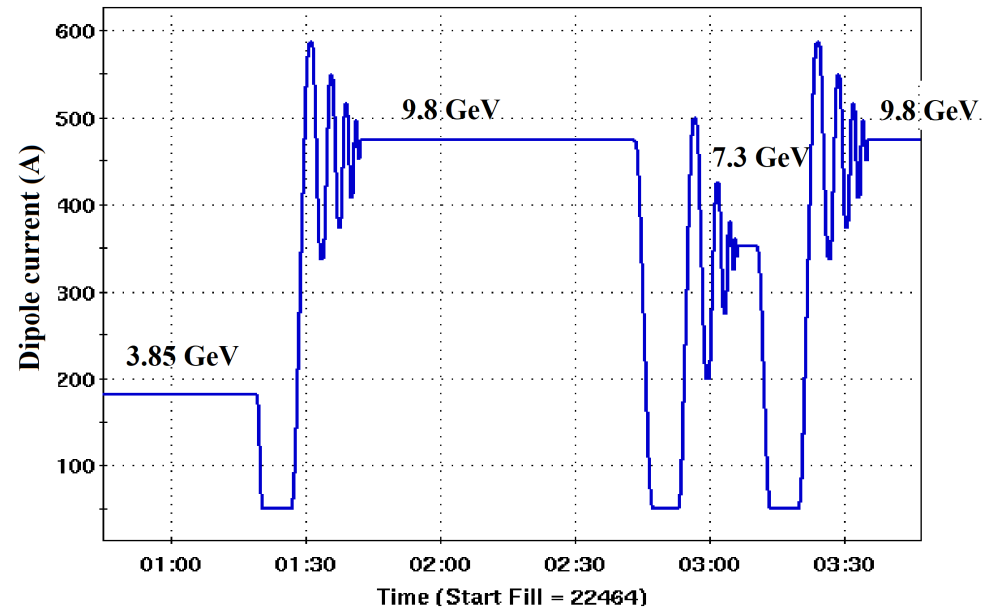
Modes in Run-19



Electron cooling planned

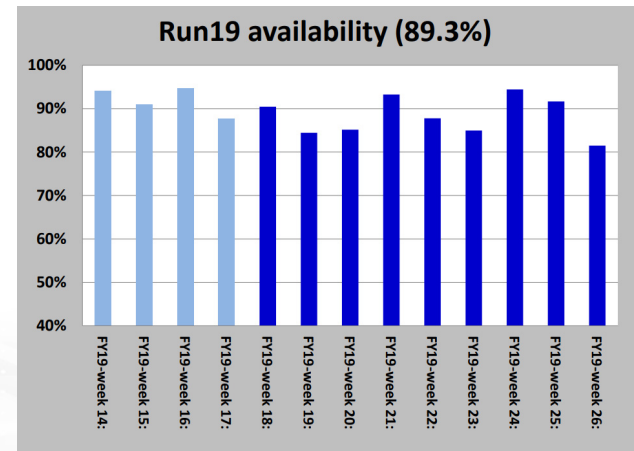
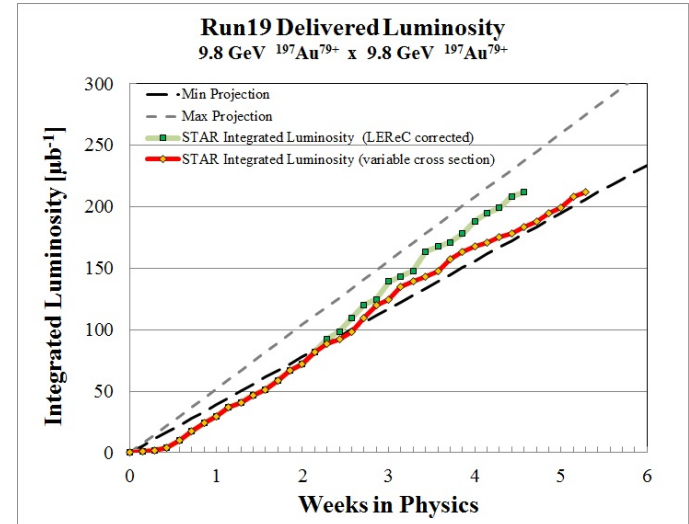
Mode-switching and wiggles

- RHIC switches between LEReC cooling commissioning and physics mode (different energies) almost everyday.
- Quickly switch between energies, zero wait time for persistent currents to settle down.
- Automatically restore correct settings for systems.
- Orbit, tune and chromaticity stability much improved.



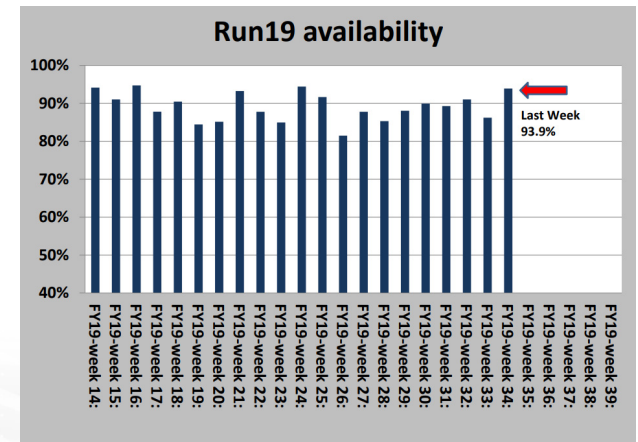
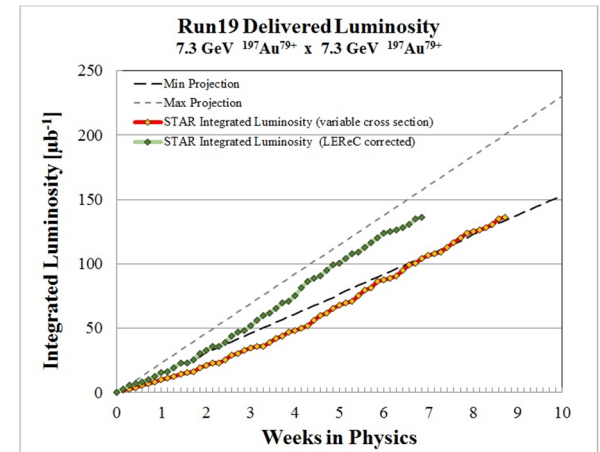
Performance at 9.8 GeV/nucleon

Parameters	Unit	Value	Note
Bunch intensity	ppb	1.8E9	Bunched, at collision, limited by beam-gas
Beta star	m	2	Started with 3
Working point		(0.09, 0.085)	
f_RF	MHz	28	
Setup time	days	7	Include other energies
Time between stores	minutes	20	
Store length	minutes	60	
Peak luminosity	$10^{24}\text{cm}^{-2}\text{s}^{-1}$	500	
Average luminosity	$10^{24}\text{cm}^{-2}\text{s}^{-1}$	150	



Performance at 7.3 GeV/nucleon

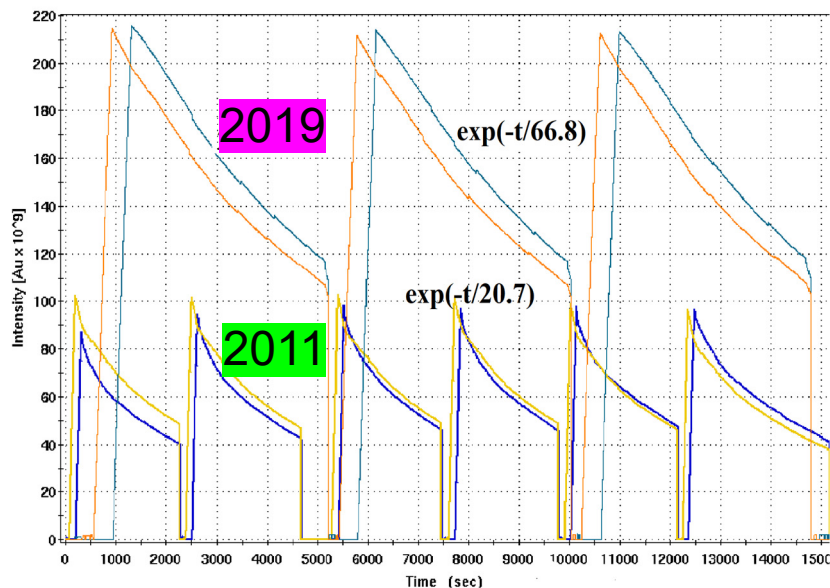
Parameters	Unit	Value	Note
Bunch intensity	ppb	1.8E9	Bunched, at collision, limited by bucket area
Beta star	m	B: 2.5, Y: 3	Comprise bwt lumi versus background
Working point		(0.09, 0.085)	Low loss with collision
Setup time	days	1	AGS RHIC synchro
f _{RF}	MHz	28	
Time between stores	minutes	20	
Store length	minutes	45	
Peak luminosity	$10^{24}\text{cm}^{-2}\text{s}^{-1}$	240	
Average luminosity	$10^{24}\text{cm}^{-2}\text{s}^{-1}$	60	



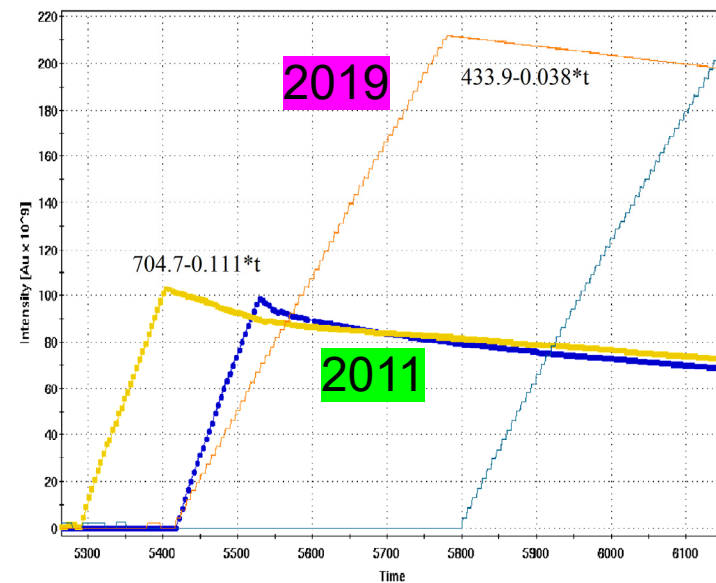
Comparison at 9.8 GeV/nucleon

2011: Two collision points, 2.5/3 m beta star at IP6&8, regular hysteresis, gap cleaning

2019: One collision point, 2 m beta star at IP6, wiggly hysteresis



Better lifetime

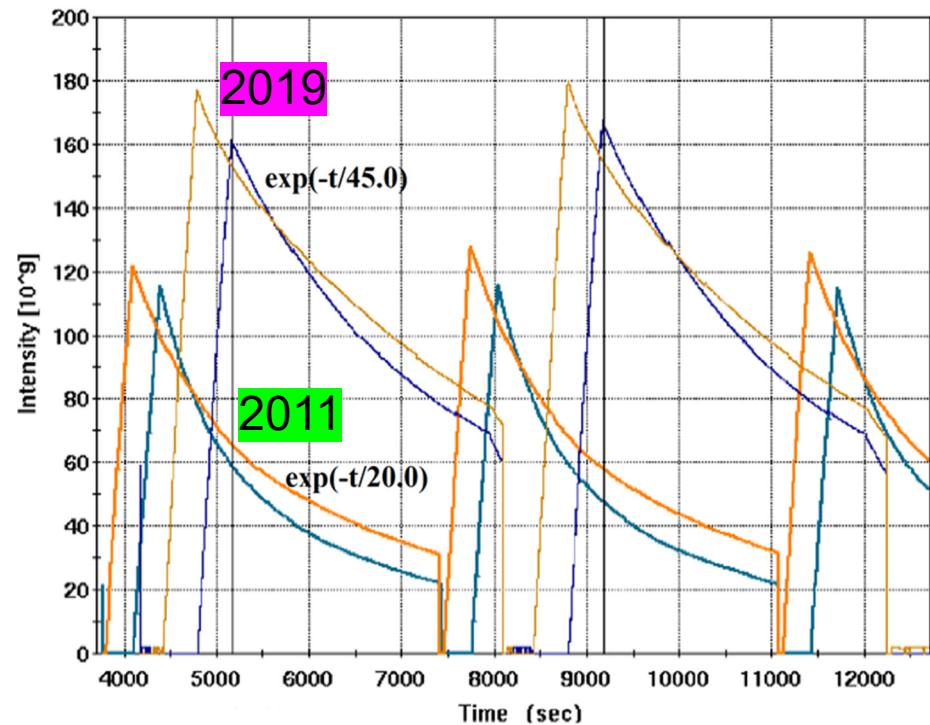


lower loss when injecting the other beam

Comparison at 7.3GeV

Two major changes in RHIC contributed to higher intensity and lower loss rate:

- Lattice design
- Demagnetization cycle

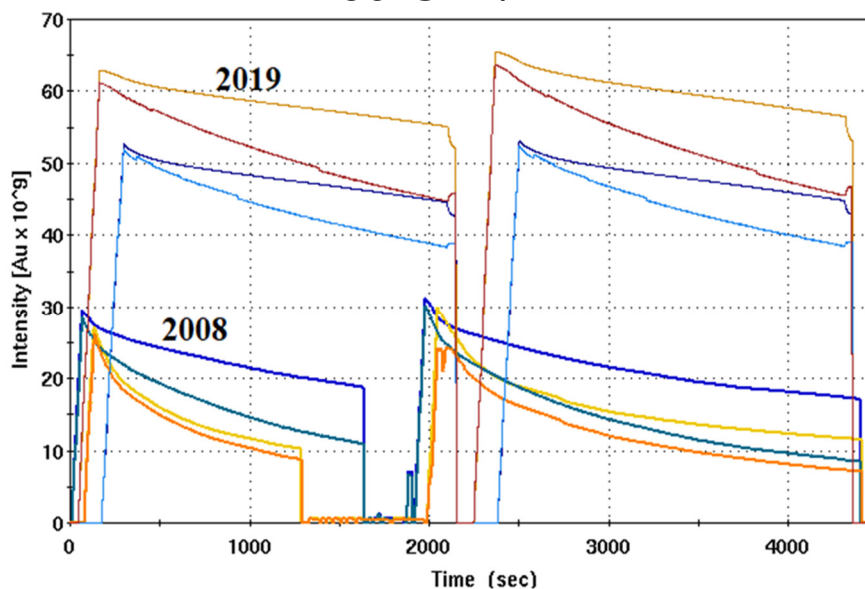


Comparison at 4.59 and 3.85 GeV/nucleon

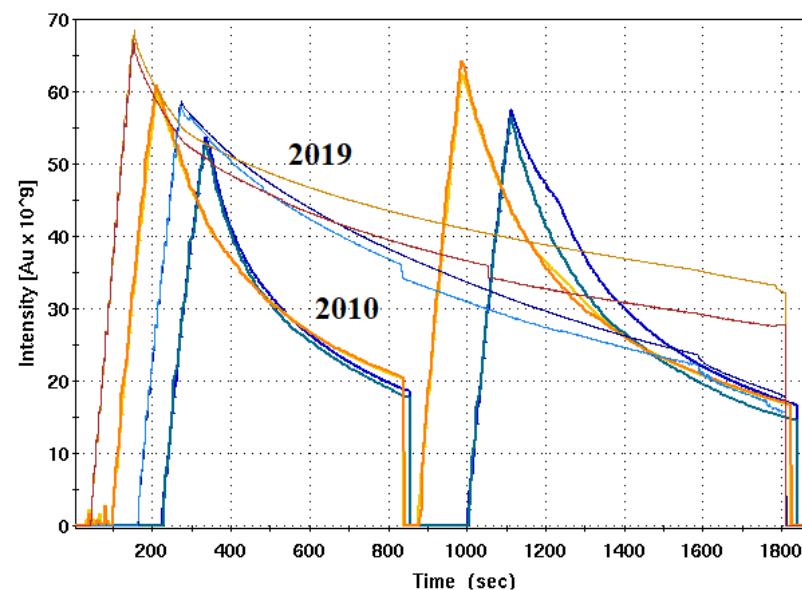
Higher bunch intensity in 2019

Better lifetime mainly due to longer bunches with 9 MHz cavity

4.59 GeV/nucleon

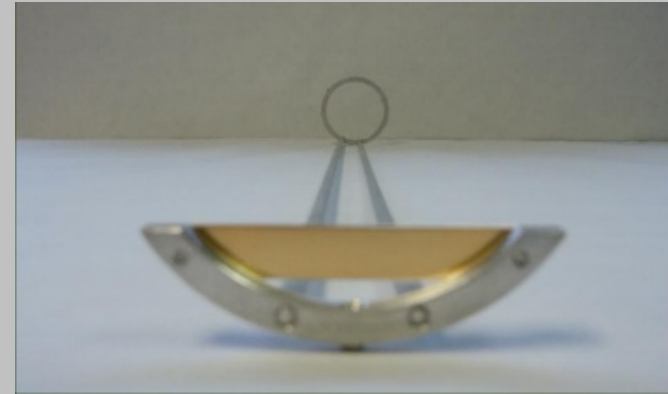


3.85 GeV/nucleon



Motivation of fixed target experiments

- It is extremely difficult to get reasonable event rate at lowest energy in colliding mode. The FXT program will allow us to extend BES-II down to $\sqrt{s} = 3.0$ GeV.
- The FXT program will allow for comparison with collider mode analyses at the same energies.
- The target foil is held 2 cm below of the beam axis. The foil is 1 mm thick.



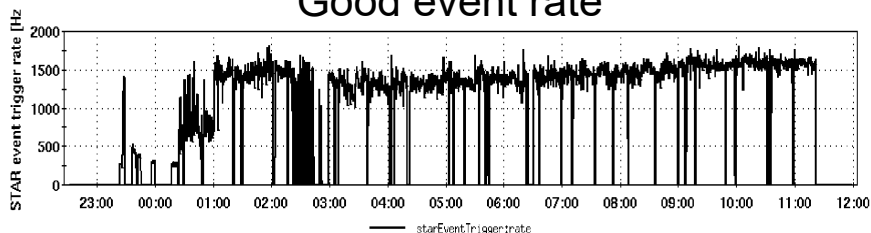
beam energy [GeV/n]	$\sqrt{s_{NN}}$ [GeV/n]	run duration [days]	no. of even
9.8	19.6	10	36 M
7.3	14.6	24	20 M
5.75	11.5	10	12 M
3.85	7.7	32	4 M
2.5	5.0	0.2	1

Fixed target runs in 2019

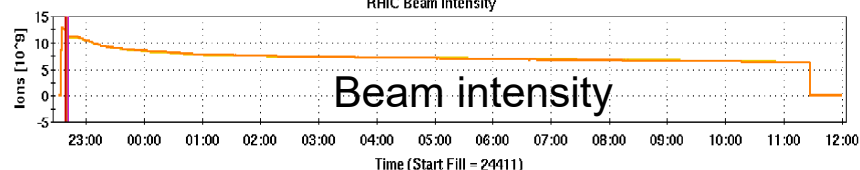
- Larger beta-function (10 m) at STAR experiment is very helpful for controlling the rates.
- Background was well under controlled.
- The base-band tune meter kickers and orbit control are the two primary ways to control STAR event trigger rate. Beam size at the fixed target location was too small at beam energy 31.2 GeV and the diffusion rate was too low. Yellow horizontal chromaticity was reduced to blow up emittance to get reasonable rate.
- For future fixed target runs at high energy, the following options can be explored: large beta function, tune meter kicker, adjustment of chroms and tunes.

FXT at 31.2 GeV ($\sqrt{s}=7.7$ GeV)

Good event rate

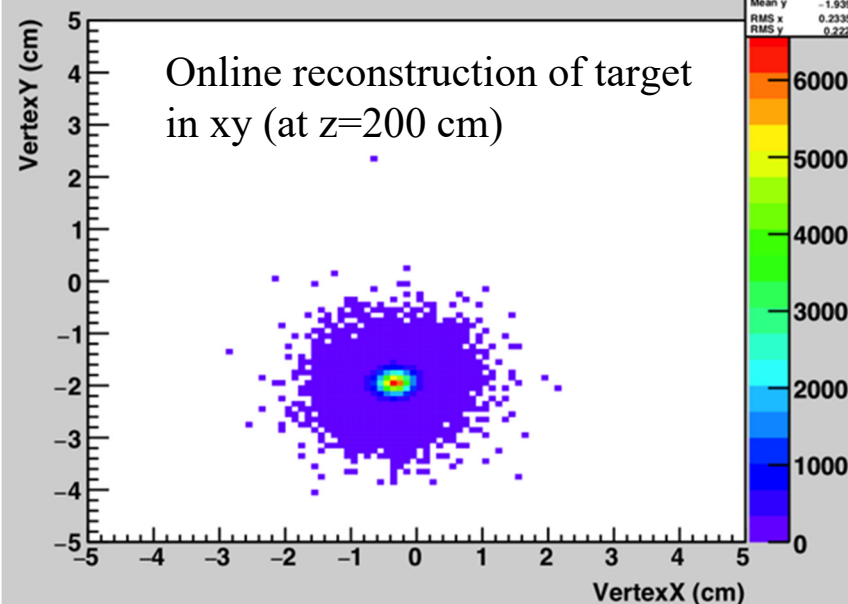


RHIC Beam Intensity



Beam intensity

VertexXY



Summary

- RHIC physics running and LEReC commissioning coexisted near perfectly during 2019.
- RHIC performance at 9.8 and 7.3 GeV were substantially improved with new developments compared to previous running.
- LEReC team achieved their tremendous goals during cooling commissioning in 2019, and demonstrated for the first time cooling of ion bunches with RF accelerated electron bunches.
- Fixed target experiments running conditions were improved with larger beta star. Future FXTs will benefit as well.
- We are planning for a challenging RHIC running with operational LEReC next year.

**Fixed target operation: WEPLH08 (poster),
TUPLO05 (poster)**

**LEReC cooling: TUZBA1 (talk), TUZBB3 (talk),
THZBA5 (talk), Poster TUPLM06 (poster)**

Thank you for your attention!

What's new compared to Run-11

For physics at 9.8 and 7.3 GeV/nucleon:

- Wiggle (demagnetization) ramp for RHIC superconducting magnets to reduce persistent current effects.
- New betatron working point at 0.09/0.08.
- Beta squeeze at IP6 for higher luminosity.
- Upgrade of corrector control from 12 to 16 bit.

For LEReC cooling commissioning:

- New 9 MHz cavities for 3.85 GeV/nucleon, much improved lifetime.
- Improved lifetime due to wiggle (demagnetization) ramp.

Operational challenges for physics at 9.8 and 7.3 GeV

- Beam-gas interaction induced background, beam scrubbing was not quite effective at 9.8 GeV.
- Limited bucket acceptance at 7.3 GeV: tight tolerance on injection matching, more de-bunched beam.
- Background control: injection kicker kicking de-bunched beam; beam scraping in the triplets with squeezed beta star.
- Run gap cleaning at the end of stores to avoid generating background and hurting bunched beam.